A Conditional Diffusion-Based Crack Segmentation Model: ConditionCrack Segmentation

Jihe Xu, Chengtao Zhang, and Wei Zeng

***Abstract*— Pavement cracks pose a direct threat to the safety of transportation systems. ConditionCrack Segmentation (CCS), a pavement crack segmentation model based on conditional diffusion, is proposed in this study. The model progressively removes noise from a random Gaussian distribution, and it is guided by image conditions to generate crack predictions. The model comprises two components: the image encoder and the map decoder. As the image encoder, the combination of MiT and FPN, aggregates multi-scale features and uses the encoded feature map as a conditional input to supplement and constrain noised labels. The map decoder uses stacked transformer layers, reducing the number of parameters in the denoising network and enhancing the ability of the decoder to model both the local details of fractures and global sequences. The model introduces Classifier-Free Guidance (CFG) and a Sampling Drift Calibration Module (SDCM) to adaptively calibrate the decoder, ensuring the effectiveness of the conditional inputs. The proposed CCS approach achieves the highest mIoU on all three evaluated public datasets (mIoU = 84.46 on CRACK500, mIoU = 89.17 on DeepCrack, and mIoU = 79.47 on CFD), outperforming both mainstream semantic segmentation models and pavement crack segmentation models using diffusion models. Moreover, the proposed CCS model achieves 32 FPS, providing a practical real-time detection system framework after performing structural optimization.**

***Index Terms*— Conditional diffusion model, crack detection, semantic segmentation, transformer, aggregated pixel attention.**

I. Introduction

WITH the continuous advancement of road infrastructure and the development of the automotive industry, China's total road mileage and vehicle ownership rate have grown rapidly. By the end of 2023, China's road network is expected to cover 5,441,000 kilometers, with vehicle ownership surpassing 435 million. This trend makes road safety issues increasingly prominent. Pavement cracks are the most prevalent type of roadway distress, and they pose a direct threat to road safety. Under high-speed driving conditions, road surface damage can cause vehicles to bump or lose control, potentially resulting in serious accidents. Therefore, detecting and repairing pavement cracks in a timely manner is crucial. Delayed crack repair [1] often increases both the complexity and cost of road maintenance. As autonomous driving technology advances, vehicles must detect and identify potential hazards to ensure safe driving. Additionally, to reduce the burden imposed by road maintenance, it is necessary to detect and repair cracks at an early stage. This not only involves the safety of individual vehicles but also serves as an important guarantee for the safe operation of the entire transportation system.

With the advancements achieved in deep learning, pavement crack detection has been extensively implemented using these techniques [2], [3]. Object detection models typically output crack recognition results as regions selected as anchor boxes [4]. However, since the spatial distributions of cracks are usually continuous and inhomogeneous, these models often segment the target into multiple discontinuous regions. In contrast, semantic segmentation models such as U-Net [5], FCN [6], SegFormer [7], and DeepLabV3+ [8] generate pixel-level classification results, accurately labeling the shape and location of cracks. In recent years, researchers have proposed specialized semantic segmentation frameworks, such as the lightweight SDDNet [9], STRNet [10] and UNet-VGG19 [11] et al. models, to address the characteristics of road crack structures, including their weak contrast levels and complex background interference. This fine-grained segmentation strategy is crucial for accurately understanding and effectively addressing pavement cracks.

Recently, diffusion model [12] has been extensively applied in image generation tasks. Their semantic segmentation applications, particularly in medical imaging [13], has been highly successful. This type of model generates data by gradually adding noise, transforming them from their original state to random noise, and then reversing the process. This process is considered a parameterized Markov chain [14]. For semantic segmentation, the diffusion model is trained to predict the pixel-level noise at each step, characterizing the joint probability distribution of all pixels. This approach allows the model to learn a broader range of spatial relationships, capturing more semantic and geometric information. It can adapt to the elongated and irregular structures of cracks, which makes the model more robust to small crack variations, enabling efficient background separation and optimized feature representation.

Received 28 October 2024; revised 9 March 2025 and 25 April 2025; accepted 28 May 2025. Date of publication 19 June 2025; date of current version 21 October 2025. This work was supported in part by the Central Guidance for Local Scientific and Technological Development Funds under Grant Guike ZY23055014 and in part by the National Natural Science Foundation of China under Grant 52202491. The Associate Editor for this article was Y. Yu. *(Corresponding author: Chengtao Zhang.)*

Jihe Xu is with the School of Mechanical and Automotive Engineering, Guangxi University of Science and Technology, Liuzhou 545006, China (e-mail: 20230104011@stdmail.gxust.edu.cn).

Chengtao Zhang is with the School of Mechanical and Automotive Engineering, Guangxi University of Science and Technology, Liuzhou 545006, China, and also with the Engineering Research Center for Advanced Design and Manufacturing of Heavy Vehicle Components, Ministry of Education, Liuzhou 545006, China (e-mail: zhchtzct@gxust.edu.cn).

Wei Zeng is with the School of Mechanical and Automotive Engineering, Guangxi University of Science and Technology, Liuzhou 545006, China, and also with the Research Center of Intelligent Equipment, Beijing Academy of Agriculture and Forestry Sciences, Beijing 100097, China (e-mail: 20230102003@stdmail.gxust.edu.cn).

Digital Object Identifier 10.1109/TITS.2025.3577492

1558-0016 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Civil Aviation University of China. Downloaded on February 07,2026 at 12:25:49 UTC from IEEE Xplore. Restrictions apply.

Previous semantic segmentation networks based on diffusion models require gradual diffusion during the inference process. Each step involves multiple inferences for conducting feature extraction, significantly increasing the computational overhead as the number of diffusion steps increases [15].

It has been shown that the simple integration of diffusion models into semantic segmentation tasks is not reasonable. The object to which noise is added is more important than the type of noise used during training [16]. The traditional semantic segmentation classifies pixel of the input image into the pixel of the corresponding label by learning the features of the real image, i.e., maximizing the posterior. Utilizing the semantic constraints imposed by the segmentation labels (the morphology of the continuous crack space) as prior probabilities for diffusion, it is reasonable to employ these labels as targets for noise addition. By using the prior probabilities, the initial sampling drift issue can be overcome by maximizing the prior distribution. However, the traditional labeling methods lead to misclassified pixels (e.g., classifying stones in cracks as cracks). Additionally, owing to the extreme imbalance in crack datasets, where more than 90% of pixels are background pixels [17], the utilized model often classifies more details as background information to achieve convergence, leading to erroneous segmentation results.

In pavement crack segmentation tasks, the image generation capabilities of the diffusion model can significantly compensate for misclassification because they neglect labels and errors during feature extraction. However, the diffusion model is not highly controllable and can only generate images randomly based on the basis of the learned distribution [18]. This causes the quality of the training images to have a significant effect on the final detection results. To address this issue, introducing conditional diffusion modeling is essential. A conditional diffusion model can incorporate specific conditional variables (e.g., text and images) into the generation process, thus making the generated images more compatible with specific needs and conditions. Conditioning on image features makes the segmentation process controllable, restoring the lost label details and correctly identifying cracks that are often categorized as background information. This approach balances the segmentation results between the true values and the labels.

To address these issues, we propose a pavement crack segmentation model that introduces a conditional diffusion-based generation model (called CCS in this paper). The key contributions of this research are summarized as follows:

1) **Conditional guidance diffusion segmentation framework:** A conditional diffusion segmentation framework with an encoder-decoder structure is constructed. The multi-scale features extracted by the image encoder are used as conditional inputs and are concatenated with labels possessing Gaussian with noise different intensities. This framework constrains the randomness of the diffusion process within the conditional generation space, achieving a deterministic mapping from the noise to segmentation results through a joint label diffusion and conditional guidance Mechanism. Compared with traditional step-by-step diffusion methods, it consumes less computational resources and significantly improves the segmentation quality of semantic boundaries.
2) **Dual-optimization training:** A collaborative optimization strategy is designed with Classifier-Free Guidance (CFG) and a Sampling Drift Calibration Module (SDCM). The CFG mechanism ensures the availability of a conditional input while establishing an adaptive balance between label confidence and data distribution. The SDCM constructs an innovative prediction–annotation dual-feedback loop that calibrates the noise distribution through feature space alignment.
3) **Optimized stacking map decoder:** A novel map decoder with stacked transformer layers with Aggregated Pixel Attention (APA) mechanism and depthwise separable convolutional Feed-Forward Networks is proposed. Pixel-level long-range dependency modeling is achieved through APA mechanisms. This architecture simultaneously enhances the denoising capabilities of the decoder and reinforces the integration between local crack features and global contextual information. Compared with conventional approaches, the proposed design achieves more precise and robust semantic segmentation effects while requiring fewer parameters.
4) **Comprehensive performance evaluation:** Training and validation processes are conducted on the CRACK500, DeepCrack, and CFD datasets to verify the superior performance of the proposed diffusion model in crack segmentation tasks. Large-parameter and lightweight models are designed to reveal the dynamic adaptation law between parameters quantity and the complexity of data. The lightweight model achieves a running speed of 32FPS with a slight reduction in Recall metric. These results establish a viable methodology for optimizing the efficiency-accuracy tradeoff through architectural refinement while preserving the fundamental advantages of diffusion-based generative modeling in crack segmentation tasks.

The rest of this paper is organized as follows. Section II presents a brief overview of the existing research on crack detection and diffusion model-based semantic segmentation methods. Section III details the principles of the produced diffusion model, conditional diffusion model, and CFG, SDCM, along with the principles of APA and the specific architectural design of the developed map decoder. Section IV outlines the experimental setup of CCS with fair data comparisons, analyzes the performance of our model through both quantitative and qualitative evaluations of all its modules, and discusses the complexity and lightweight performance of the model. Finally, Section V concludes the study, summarizing the main contributions and outlining potential future research directions.

II. Related Works

A. Deep Learning Based Crack Detection

Owing to the rapid development of deep learning, current feature extraction methods employed in image segmentation tasks primarily use two main types of structures: convolutional neural networks (CNNs) and transformers [19]. Model based

Authorized licensed use limited to: Civil Aviation University of China. Downloaded on February 07,2026 at 12:25:49 UTC from IEEE Xplore. Restrictions apply.

on CNNs extracts and processes local features using convolutional layers. Huyan et al. [20] proposed CrackU-Net, which outperforms the traditional U-Net model. Han et al. [21] introduced CrackW-Net, which is a residual block-free network enhanced by a skip-level round-trip sampling block to the U-Net network, and their approach produced excellent results on the CRACK500 dataset. Chen and He [22] developed AttentionCrackNet, a neural network that utilizes a VGG11-based encoder-decoder architecture with skip connections and attention gates to precisely and efficiently detect road cracks at the pixel level. Wang et al. [23] utilized YOLOX [24] and a dual self-attention segmentation network to detect and segment road potholes. Although techniques such as dilated convolution and attention mechanisms have been integrated into CNN models, CNN-based crack detection techniques still face limitations because of their local receptive fields. This results in an insufficient ability to learn the spatial continuity of cracks and a tendency to detect discontinuities. Models based on vision transformers (ViTs) [25] extract and integrate global contextual information by using a self-attention mechanism. Guo et al. [26] proposed CTv2, a crack transformer architecture with a two-stage separation and combination strategy, to improve the efficiency and accuracy of pavement crack detection in engineering practice. Liu et al. [27] used a transformer coding block with relative positional embedding and long-distance interactions. They employed feature fusion by using a scaled-attention fusion module to highlight semantic features and suppress nonsemantic features. Tao et al. [28] designed two modules for boundary sensing of cracks to improve the local details and boundary information of cracks, and good results were achieved on both the CRACK500 and DeepCrack with a smaller number of parameters. Qu et al. [29] introduced a multi-scale convolutional feature fusion module that integrates high-level features with low-level features to capture complex cracks while employing deep supervision to enhance the convergence ability and detection accuracy of the constructed model. Sun et al. [30] proposed a Deeplabv3+ based DMA-Net, which achieved impressive performance on the CRACK500, DeepCrack and FMA datasets. Both network optimization and redesign processes are crucial for improving the accuracy of pavement crack detection. However, CNN and Transformer based models lack interactions between local and global features, resulting in the loss of semantic information.

B. Semantic Segmentation Models Based on Diffusion Model

Baranchuk et al. [31] verified the excellent performance of diffusion models in semantic segmentation tasks. Diffusion models have been shown to effectively capture semantic information in input images, which is crucial for completing semantic segmentation tasks. The DDPS framework proposed by Lai et al. [16] for conducting denoising diffusion-based semantic segmentation with a priori mask modeling and the DDP framework proposed by Ji et al. [32] for implementing semantic segmentation with conditional diffusion modeling both demonstrate the superiority of denoising diffusion-based probabilistic models. Introducing a denoising diffusion model to conduct semantic segmentation yields improved performance because of the spatial and semantic properties of cracks. Han et al. [33] proposed a two-stage unsupervised and supervised semantic segmentation framework using multi-blur-based cold diffusion, which achieved an 85.8% mIoU on the CRACK500 dataset. Li et al. [34] applied a generative adversarial network (GAN) to design CrackCLF, which is an automatic generative crack detection model that uses closed-loop control for self-learning and error correction purposes. Ma et al. [35] proposed a crack detection system that integrates PCGGAN and YOLO-MF and uses GANs to generate synthetic crack images to overcome data scarcity challenges. Similarly, Ali and Cha [36] developed IDSNet, which leverages AGANs to generate internal loss ground-truth datasets to enhance training data. However, both methods do not incorporate a generative model into the process of the detection stage, limiting them to data augmentation work. Zhang et al. [15] proposed a crack diffusion model, which leverages the learning capabilities of diffusion generation models to analyze the data distributions and potential spatial relationships of cracks across different sampling time steps. This approach generates more accurate and continuous segmentation results, achieving an 84.1% mIoU on the CRACK500 and 86.2% mIoU on the DeepCrack. These studies have demonstrated the superiority of generative diffusion models in semantic segmentation tasks. However, in the field of crack semantic segmentation, the use of diffusion models is still in its infancy, with most models suffering from large parameter counts and high computational complexity levels.

III. Methodology

A. Diffusion Model

The diffusion model comprises two processes: a forward diffusion process and a reverse denoising process. The forward process involves gradually adding Gaussian noise to the input image until it becomes random noise that follows a normal distribution. Conversely, the reverse denoising process gradually reduces the noise from a Gaussian-noised image, ultimately aiming to recover the original image. The addition of noise during the forward process is defined as:

$$q(G_t|G_{t-1}) = \mathcal{N}(G_t; \sqrt{1-\beta}x_t, \sqrt{\beta}I) \tag{1}$$

By using the reparameterization trick, the final diffusion result can be obtained in one step, as shown in (2):

$$q(G_T|G_0) = \mathcal{N}(G_T; \sqrt{\bar{\alpha}}x_0, \sqrt{1-\bar{\alpha}}I) \tag{2}$$

where $\bar{\alpha}_t = \prod_{i=1}^{t}(1-\beta_i)$, β represents the noise schedule, and x_0 represents the label. The formula for the reverse denoising process is illustrated in (3):

$$q(G_{T-1}|G_T, G_0) = \mathcal{N}(G_{T-1}; \bar{\mu}(G_T, G_0), \bar{\sigma}\epsilon) \tag{3}$$

To accelerate the inference process, we employ the DDIM [37] sampling rule. DDIM accelerates the sampling procedure by modeling the diffusion process as a deterministic process. This method generates high-quality images in fewer steps,

Authorized licensed use limited to: Civil Aviation University of China. Downloaded on February 07,2026 at 12:25:49 UTC from IEEE Xplore. Restrictions apply.

thus enhancing inference efficiency. The formula for DDIM sampling is as follows:

$$G_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\left(\frac{G_t - \sqrt{1-\bar{\alpha}_t}\epsilon_\theta(G_t, t)}{\sqrt{\bar{\alpha}_t}}\right) + \sqrt{1-\bar{\alpha}_{t-1}-\sigma_t^2}\cdot\epsilon_\theta(G_t, t) \tag{4}$$

where ϵ_θ is a neural network that is trained to predict noise and $\sigma_t^2 = \eta \cdot \frac{1-\bar{\alpha}_t}{\bar{\alpha}_{t-1}}\left(\frac{1-\bar{\alpha}_t}{\bar{\alpha}_{t-1}}\right)$ is the hyperparameter that controls the balance between determinism and randomness. In deterministic sampling, η is set to 0; therefore, in this paper, we use the sampling formula shown in (5):

$$G_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\left(\frac{G_t - \sqrt{1-\bar{\alpha}_t}\epsilon_\theta(G_t, t)}{\sqrt{\bar{\alpha}_t}}\right) + \sqrt{1-\bar{\alpha}_{t-1}}\cdot\epsilon_\theta(G_t, t) \tag{5}$$

B. Conditional Diffusion Model

The diffusion model described above focuses solely on modeling the data distribution $p(x)$. The conditional diffusion model controls the data generation process through conditional information c [38]. This conditional information is incorporated, along with the timestep information, by simply adding arbitrary conditional information at each transition step. This process can be formulated as shown in (6):

$$p_\theta(G_{0:T}|c) = p(G_T)\prod_{t=1}^{T} p_\theta(G_{t-1}|G_t, c) \tag{6}$$

To ensure the availability of conditional constraints, we use Classifier-Free Guidance (CFG) [39] for performing assisted training and inference process. The CFG method involves randomly nullifying the provided conditional input during the sampling process. This enables the model to obtain both unconditional and conditional outputs, learning the variance between them to guide the sampling process. The variance is calculated as follows:

$$\nabla_x \log p_\gamma(x|y) = \nabla_x \log p(x) + \gamma(\nabla_x \log p(x|y) - \nabla_x \log p(x)) \tag{7}$$

Therefore, the model prediction process for real noise can be defined as (8):

$$\bar{\epsilon_\theta}(x_t, t, c) = \epsilon_\theta(x_t, t) + \omega[\epsilon_\theta(x_t, t, c) - \epsilon_\theta(x_t, t)] \tag{8}$$

where ω is a scaling factor. When ω is larger, the generated results align more closely with the conditional inputs, thereby reducing their diversity. As ω decreases, the diversity of the generated results increases. When ω equals 1, the predicted noise corresponds to the input conditional noise. In this paper, ω is set to 7.5. Similar to ϵ_θ in (5), ϵ_θ is a neural network that is trained to predict noise and is hereafter referred to as the map decoder.

In our setting, the data samples are $G_0 = Label$, and a neural network is trained to predict ϵ_θ from random noise $G_T \sim \mathcal{N}(0, I)$ conditioned on the corresponding image x. We choose to directly predict the ground truth, as this strategy reduces the errors accumulated during denoising, particularly in complex boundary areas, and enables end-to-end optimization with a simpler training process and better generalization across different tasks. The standard loss function for CFG is the L2 loss, which is formulated as follows:

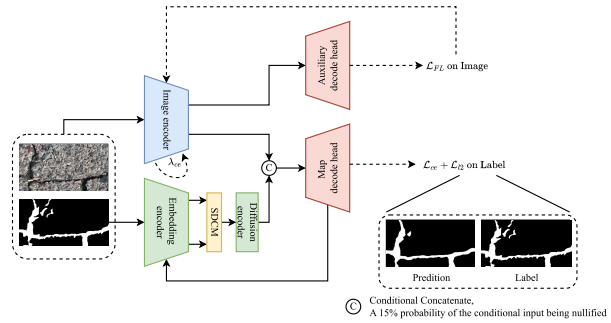

Fig. 1. Architecture of CCS.

$$l2 = \left\|\epsilon_\theta(z_\lambda, c) - \epsilon\right\|^2 \tag{9}$$

The cross-entropy loss for segmentation task is as follows:

$$celoss = -\sum_{i=1}^{size} y_i \cdot \log \hat{y}_i \tag{10}$$

C. Network Structure

This section comprehensively describes the network structure and detailed design of ConditionCrack Segmentation (CCS). In this paper, we adopt the design of DDP [32], dividing the model into two components: the image encoder and the map decoder. Fig. 1 shows the overall network architecture of CCS. We introduce CFG to enhance the imposed conditional constraints. The transformer layer of the decoder is built with Aggregated Pixel Attention and depth-separable convolution to characterize pavement cracks. The image encoder is compatible with modern backbones such as MiT [7], Swin Transformer [40].

1) Image Encoder: Compared with CNNs, transformers can more effectively utilize semantic information contained in images to establish dependencies between different pixels. In this work, we select MiT as the backbone for feature extraction, generating multi-scale features at four different resolutions. These multi-scale features are then fused using the FPN [41] neck and further aggregated by a 1×1 convolution. The resulting feature map, with a resolution of, is used as the conditional input of the map decoder. Fig. 2 illustrates the encoding process performed by the image encoder. Importantly, the image encoder is itself a neural network, and the quality of the feature extraction process significantly impacts the conditional inputs. Therefore, it is crucial to evaluate the features extracted from the backbone. The FCN [6] head is employed as an auxiliary decoder to facilitate the training procedure. Owing to the imbalance between the distributions of positive and negative crack examples, we select Focal Loss [42] as the loss function. This design reduces the duration of training while maintaining its effectiveness, and the fusion of feature maps with different sizes enables the model to capture more detailed information.

Authorized licensed use limited to: Civil Aviation University of China. Downloaded on February 07,2026 at 12:25:49 UTC from IEEE Xplore. Restrictions apply.

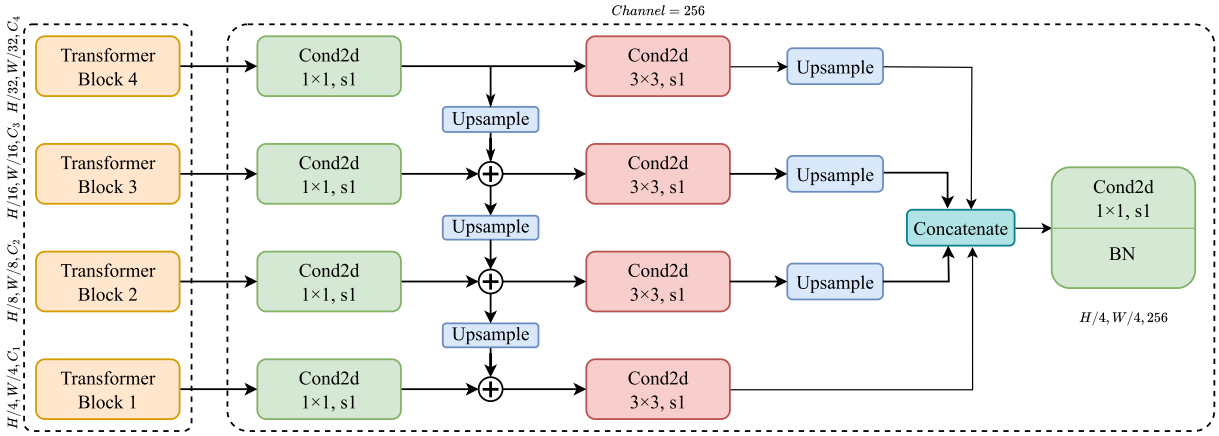

Fig. 2. The process of image encoder.

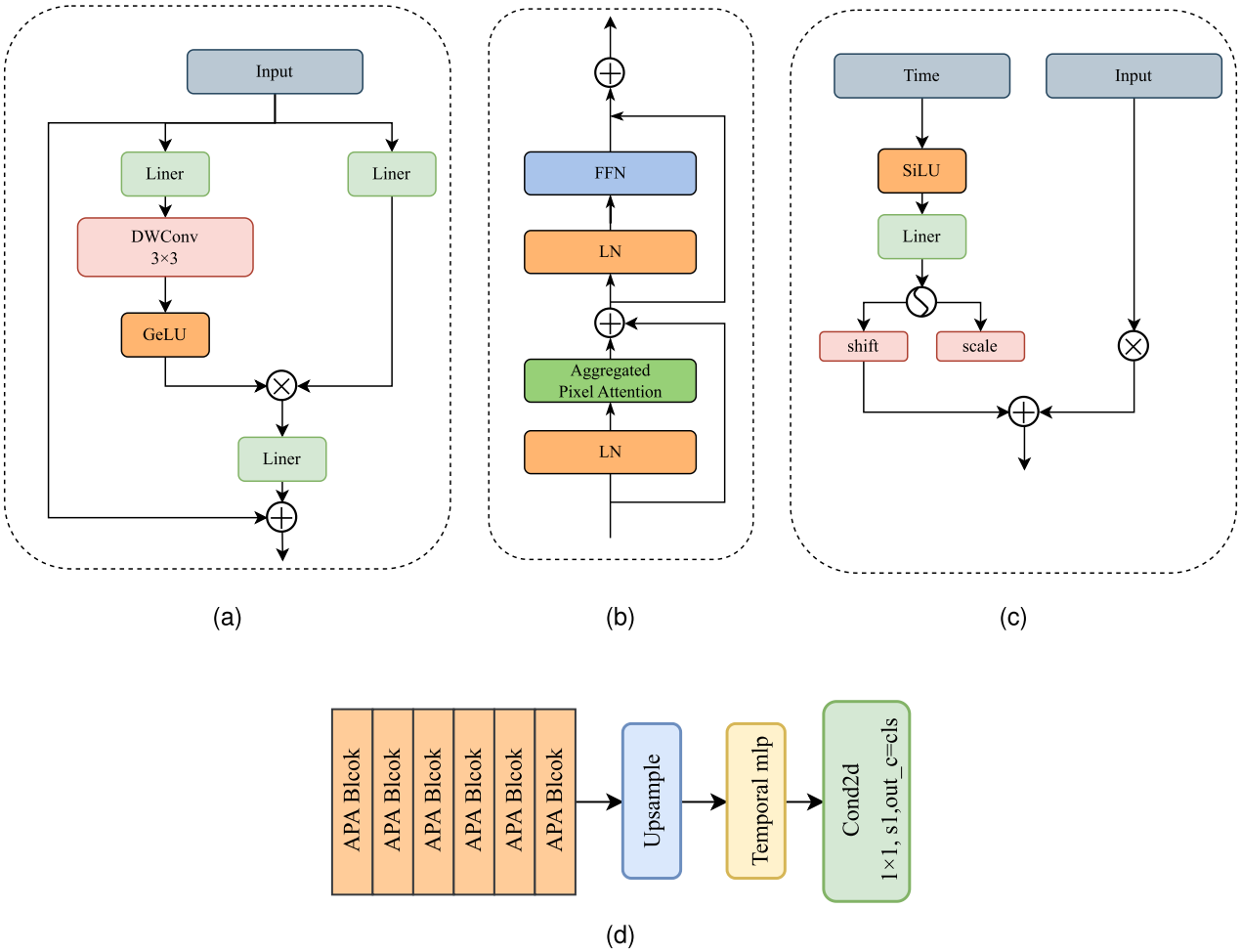

Fig. 3. The structures of (a) FFN layer, (b) Transformer layer, (c) Temporal mlp and (d) Map decoder.

2) Map Decoder: The encoding and decoding parts of previous diffusion models employed U-Net or U-Net like structures [13], [43]. In this paper, we use a learnable stacked transformer layer as the decoder. The transformer layer follows the structure of ViT and is composed of normalization, self-attention, normalization, and FFN layers. To achieve temporal embedding, we design a temporal mlp. With the input diffusion time step, scaling factors and shifts are obtained. The prediction is then multiplied by the scaling factors and adjusted by adding the shift, resulting in the prediction for the embedded time step. We employ Aggregated Pixel Attention as the self-attention layer. This design reduces the number of redundant computations, making the map decoder lightweight and compact while improving the stability and efficiency of the training process. The structures of the transformer block, FFN layer, temporal mlp and map decoder are shown in Fig. 3.

Authorized licensed use limited to: Civil Aviation University of China. Downloaded on February 07,2026 at 12:25:49 UTC from IEEE Xplore. Restrictions apply.

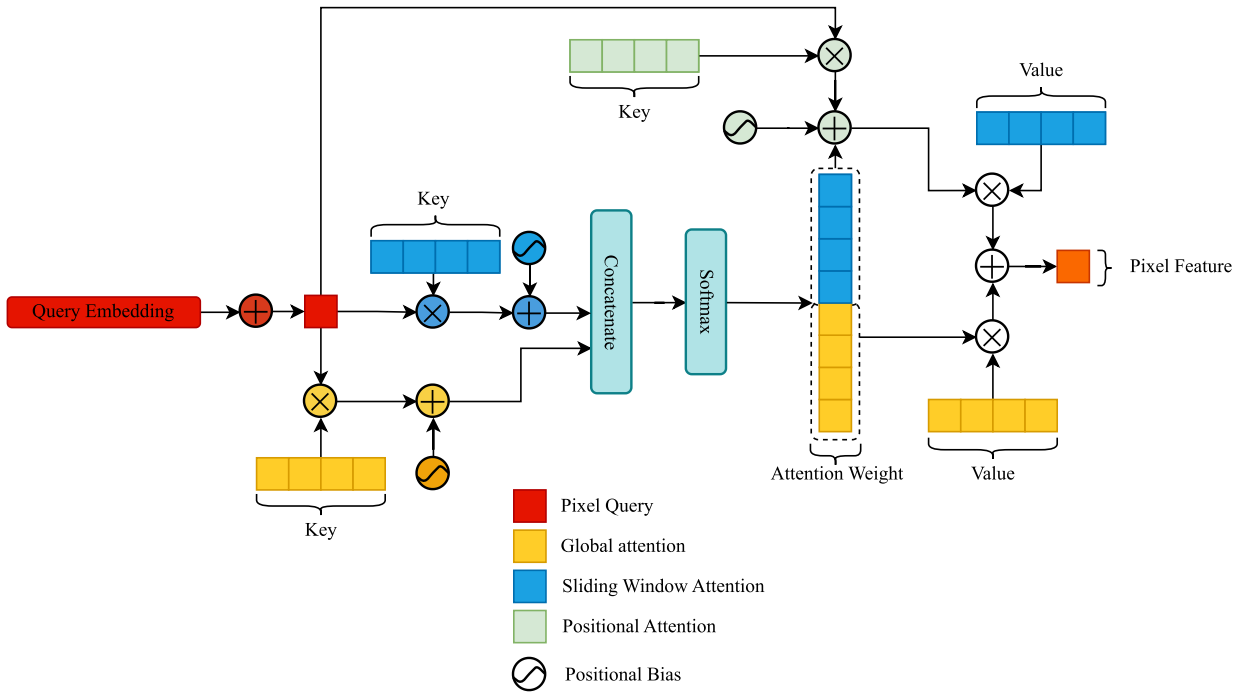

Fig. 4. Main structure of aggregated pixel attention.

3) Aggregated Pixel Attention: To link the semantic information contained in the local details of cracks with the global morphology, we follow Shi's [44] research in TransNeXt; the pixel-focused attention mechanism incorporates query-centered sliding window attention and pooling attention for multiple local and global information sensing operations. Concatenating the keys derived from the global and sliding window attention and using the Softmax function to compute the similarity between their query keys facilitates the mutual learning of local and global features, leading to improved feature learning outcomes. For query embedding, we use the pixels of the input image as queries to perform cross-attention weight calculations. To learn about pixel and global positional offsets, we integrate a set of learnable tokens $T \in \mathbb{R}^{d\times k^2}$ into each attention head to obtain an additional dynamic position bias and add it to query keys. The pixel-focused attention can be formulated as (11):

$$PFA(X_{(i,j)}) = (A_{(i,j)\sim p(i,j)} + Q_{(i,j)}T)V_{p(i,j)} + A_{(i,j)\sim\sigma(X)}V_{\sigma(X)} \tag{11}$$

The use of cosine similarity to generate more appropriate attention weights can effectively enhance the stability of the model training process [45]. Following Su's [46] study, we propose length-scaled cosine attention as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}(\tau \log N * \hat{Q}\hat{K}^T)V \tag{12}$$

where N denotes the number of effective keys that each query interacts with, τ is the learnable variable for each attention head with an initial value of $\frac{1}{0.24}$, and we define the queries and keys with $l2$ normalization applied along their head dimensions as $\hat{Q}$ and $\hat{K}$, respectively. We apply log-spaced continuous position bias (log-CPB) [47] on the pooling feature path. Combined with the above design, the APA method can be expressed as follows:

$$S_{(i,j)\sim p(i,j)} = (\hat{Q}_{(i,j)} + QE)\hat{K}^T_{p(i,j)}$$
$$S_{(i,j)\sim\sigma(i,j)} = (\hat{Q}_{(i,j)} + QE)\hat{K}^T_{\sigma(X)} \tag{13}$$
$$B(i,j) = Concat(B_{(i,j)\sim p(i,j)}, \text{log-CPB}(\Delta_{(i,j)\sim\sigma(X)})) \tag{14}$$
$$A(i,j) = \text{softmax}(\tau \log N * \text{Concat}(S_{(i,j)\sim p(i,j)}, S_{(i,j)\sim\sigma(X)}) + B_{(i,j)}) \tag{15}$$
$$A_{(i,j)\sim p(i,j)}, A_{(i,j)\sim\sigma(X)} = \text{Split}(A_{(i,j)}) \text{ with size } [k^2, H_p, W_p] \tag{16}$$
$$\text{APA}(X_{(i,j)}) = (A_{(i,j)\sim p(i,j)} + \hat{Q}_{(i,j)}T)V_{p(i,j)} + A_{(i,j)\sim\sigma(X)}V_{\sigma(X)} \tag{17}$$

Fig. 4 illustrates the main structure of the APA mechanism.

4) Sampling Drift Calibration Module: Previous studies have indicated that diffusion models frequently experience sampling drift [48] during the inference process. As the utilized model denoises imperfect predictions, it impairs the ability to detect latent semantic information, and the use of fewer sampling steps worsens this issue. The sampling drift problem becomes more severe when the sample distribution significantly diverges from the ground truth. We develop the SDCM to address sampling drift. The structure of the SDCM is shown in Fig. 5.

We use the predictions output by the model as training labels. The labels are then updated by combining them with the ground truth weighted by the cosine similarity. The calculation

Authorized licensed use limited to: Civil Aviation University of China. Downloaded on February 07,2026 at 12:25:49 UTC from IEEE Xplore. Restrictions apply.

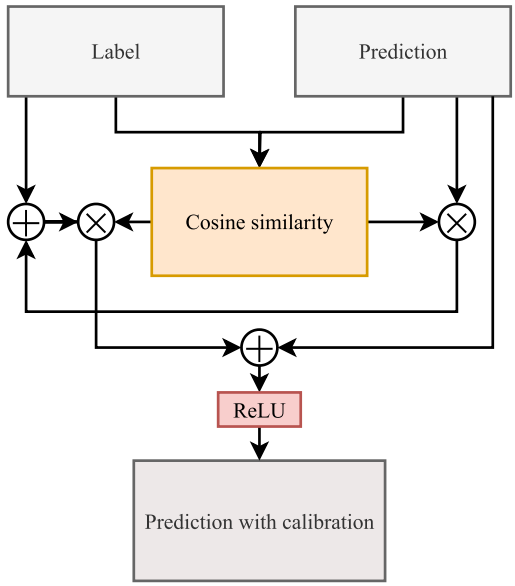

Fig. 5. Structure of SDCM.

process is shown in (18):

$$Label = RelU\left(F_a + F_b \cdot \frac{\sum_{i=1}^{c} F_{a,i} \cdot F_{b,i}}{\sqrt{\sum_{i=1}^{c}(F_{a,i}^2)} \cdot \sqrt{\sum_{i=1}^{c}(F_{b,i}^2)}}\right) \tag{18}$$

By incorporating labels into its predictions, the model can adaptively adjust its weights, accounting for the labels while preserving its original predictions.

D. Designs of the Training and Inference Processes

As described in Section III-B, we set the goal of the map encoder as the restoration of the true values instead of predicting noise. Therefore, the map encoder is actually represented as $\hat{x}_\theta(x_t, t, y) \approx x_0$. We use a form of cosine noise [14] to obtain the signal-to-noise ratio (SNR) and thus control the noise level, as shown in (2). When adding a conditional input, we set the probability of the condition being null to 15%. This design allows training a single neural network to infer denoising results both conditionally and unconditionally. The standard diffusion model assumes that the input data are continuous, but the labels used for semantic segmentation tasks are typically discretized as pixel-by-pixel classifications; thus, it is necessary to perform label encoding [49]. We use class embedding, which employs a learnable embedding layer to project discrete labels into a high-dimensional continuous space, with a sigmoid function applied for normalization. The structure of the label encoding and diffusion module is shown in Fig. 6. The formula can be expressed as follows:

$$\sigma(x) = \left(\frac{1}{1+e^{-x}} \times 2 - 1\right) * \lambda \tag{19}$$

where λ is the scaling factor, which is set to 0.01 in this paper. We use the DDIM sampling rule with conditional control via CFG and set the sampling step size to 2. Following [50], we use asymmetric time intervals during the inference stage

Algorithm 1 Training

Input: images, labels and time
  Repeat
    Encode images as features with backbone
    Sample log SNR λ
    Get random noise $\epsilon \in \mathcal{N}(0, 1)$
    Get x from the map decoder (with no grad)
    $x \sim SDCM(label, x)$
    Encode label and x with (19)
    $z_\lambda \sim (x, c)$, where c is possibly unconditional
    Corrupt labels to sampled log SNR value with (2)
    Update $loss$ with (20)
  Until the iterative process stops

Algorithm 2 Sampling

Input: encoded images x and time steps T
  Get Noise $G_T \sim \mathcal{N}(0, I)$
  for each pair of $(t-1, t)$ in T **do**
    Get input $G_t^c \sim cat(x, G_t)$, $G_t^\emptyset = G_t$
    Get α_{t-1}, α_t from log SNR λ_{t-1} and λ_t
    $G_{t-1}^c, G_{t-1}^\emptyset \sim \bar{\epsilon_\theta}(G_t^c, G_t^\emptyset, t)$
    $N_{t-1}^c \sim \frac{G_t^c - \sqrt{1-\alpha_{t-1}} G_{t-1}^c}{\sqrt{\alpha_{t-1}}}$
    $N_{t-1}^\emptyset \sim \frac{G_t^\emptyset - \sqrt{1-\alpha_{t-1}} G_{t-1}^\emptyset}{\sqrt{\alpha_{t-1}}}$
    $N_{t-1} \sim CFG(N_{t-1}^c, N_{t-1}^\emptyset)$ with (8)
    $G_t \sim \sqrt{1-\alpha_t} G_{t-1}^c + \sqrt{\alpha_t} N_{t-1}$
  end for
  return G_{t-1}^c

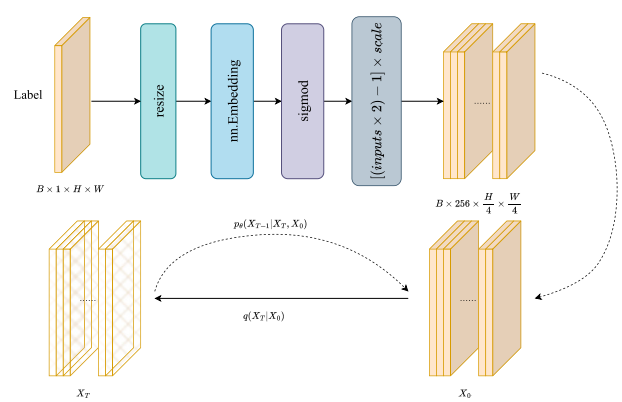

Fig. 6. Structure of label encoding and diffusion module.

and set them to 1. Following Section III-B and III-C, the loss function of this work is shown in (20):

$$loss = \lambda_1 CrossEntropyLoss + \lambda_2 l2 + \lambda_3 Fl \tag{20}$$

where λ_1, λ_2, and λ_3 represent the weights of the three losses. λ_1 is set to 1.0, λ_2 is set to 4.0, and λ_3 is set to 0.4. Algorithms 1 and 2 present the pseudocodes for the training and inference processes, respectively.

IV. Experiments and Discussions

A. System Configuration and Training Implementation

The PyTorch library version 1.13.1 is used for model training. The graphics workstation is equipped with 128 GB of

Authorized licensed use limited to: Civil Aviation University of China. Downloaded on February 07,2026 at 12:25:49 UTC from IEEE Xplore. Restrictions apply.

memory, two Intel® Xeon(R) Gold 6248R CPUs @ 3.00 GHz, an NVIDIA RTX 3090 GPU with 24 GB of VRAM, and the Ubuntu 22.04 LTS. All the models are constructed via MMSegmentation [51]. We optimize our CCS models using the AdamW [50] optimizer, with an initial learning rate of 0.00006 and a weight decay rate of 0.01, by employing a polynomial decay strategy to adjust the learning rate and a linear warm-up process. All the models are trained with a batch size of 4 over 80k iterations and evaluated every 8,000 iterations.

To fully validate the performance of the proposed model, we select three public datasets: CRACK500 [52], DeepCrack [53], and CFD [54] for training and validation. The data augmentation procedure consists of the following steps: (1) Resize the input image to a random size between 0.5 and 2 times its original dimensions. (2) Extract a 256×256 pixel patch from the scaled image, ensuring that the region of interest (ROI) containing the target class occupies no more than 75% of the cropped area. (3) Apply random flipping to the image with a probability of 50%. (4) Implement photometric distortion techniques to alter the color properties of the image. (5) Normalize the image to standardize its pixel values. (6) Pad the image to a fixed size of 256×256 pixels using a pixel value of 255.

The CRACK500 dataset contains 500 images of pavement cracks with a resolution of 2000×1500. We split these images into 640×360 segments, increasing the size of the training set to 1896 images and the size of the validation set to 348 images. DeepCrack contains 537 images containing cracks with an image size of 544×384. We select 300 images for training and 237 images for validation.

The image size of CFD is 480×320. The original 118 samples are enhanced through image inversion, contrast adjustment, and luminance changes, resulting in 565 training images and 146 validation images.

These three datasets include a variety of shooting angles, crack sizes, pavement materials, lighting conditions, weather scenarios, etc. This diversity provides comprehensive information for our model training and evaluation processes, thus enhancing the generalizability of our model.

The Precision, Recall, F1-score, and mIoU measures are commonly used evaluation metrics in crack detection and image segmentation tasks. Their formulas are as follows:

$$Precision = \frac{TP}{TP + FP} \tag{21}$$

$$Recall = \frac{TP}{TP + FN} \tag{22}$$

$$F1 = \frac{2 \times precision \times recall}{precision + recall} \tag{23}$$

$$IoU = \frac{TP}{TP + FP + FN} \tag{24}$$

Note that the number of categories used in this study is set to 2, with the background considered as one category. Consequently, the mIoU calculation includes the IoU for the background.

TABLE I
EVALUATION RESULTS OF DIFFERENT MODELS ON CRACK500

Model	Precision	Recall	F1	mIoU
U-Net	82.3	75.45	78.73	81.18
DeepLabV3+	81.02	59.31	68.49	74.34
SegFormer(b0)	81.99	76.92	79.37	81.65
CrackFormer	**84.06**	79.19	81.55	82.47
CT-crackseg	81.18	78.08	79.6	81.93
CrackDiff	81.34	84.10	81.78	84.09
DDP(Swin)	83.08	80.64	81.84	83.51
CCS(ours)	81.93	**84.29**	**83.09**	**84.46**

B. Experimental Results Analysis and Visualization Discussion

To validate the effectiveness of our models, we select the widely used and state-of-the-art (SOTA) models for comparative analyses, namely U-Net [5], DeepLabV3+ [8], and SegFormer [7], as well as the CrackFormer [27], CT-crackseg [28], as well as CrackDiff [15] and DDP [32] crack detection models, which use diffusion-generated models. All training settings are consistent with those used for CCS. Importantly, the evaluation metrics of the generative models (e.g., FID) are not the focus of this paper. Instead, we focus on exploring the ability of the conditional diffusion model to improve the semantic segmentation task.

1) Results on CRACK500: Table I shows the evaluation results for the CRACK500 dataset. The CRACK500 dataset is characterized by a short capture distance, a typical asphalt texture, and a complex surface. Among all methods, CCS and CrackDiff achieved the best results with the aid of diffusion models. CCS achieves a 0.4% higher mIoU than CrackDiff. CrackFormer, which leverages a scaling attention module, suppresses nonsemantic features while enhancing semantic features, resulting in the highest Precision.

Fig. 7 presents a visualization of the predictions produced by all methods. CCS yields the best crack continuity predictions and detects stones within the cracks. tends to predict cracks with coarser pixels and some noise points are present. This occurs because CrackDiff does not employ conditional diffusion, resulting in a less effective generation control scheme during inference and thus more random outputs and unable to accurately predict crack boundaries. DDP incorrectly classifies pixels surrounding the cracks as cracks, leading to coarser predictions. This is because while DDP also uses conditional diffusion, its map decoder uses only deformable attention, which leads to poor fine-grained prediction results. CT-CrackSeg's crack morphology predictions are highly competitive because of the integration of boundary masks during the training process, resulting in superior crack boundary predictions.

2) Results on DeepCrack: Table II presents the evaluation results for the DeepCrack dataset. Although the cracks in DeepCrack are diverse, most of them exhibit clear features with minimal interfering elements. All algorithms perform effectively on the DeepCrack dataset. CCS achieves the highest mIoU, surpassing that of CrackDiff by 2.93%. This is attributed to the use of APA by CCS, which enhances the integration of global and local semantic information. The high Precision (95.53) and low Recall (63.2) values

Authorized licensed use limited to: Civil Aviation University of China. Downloaded on February 07,2026 at 12:25:49 UTC from IEEE Xplore. Restrictions apply.

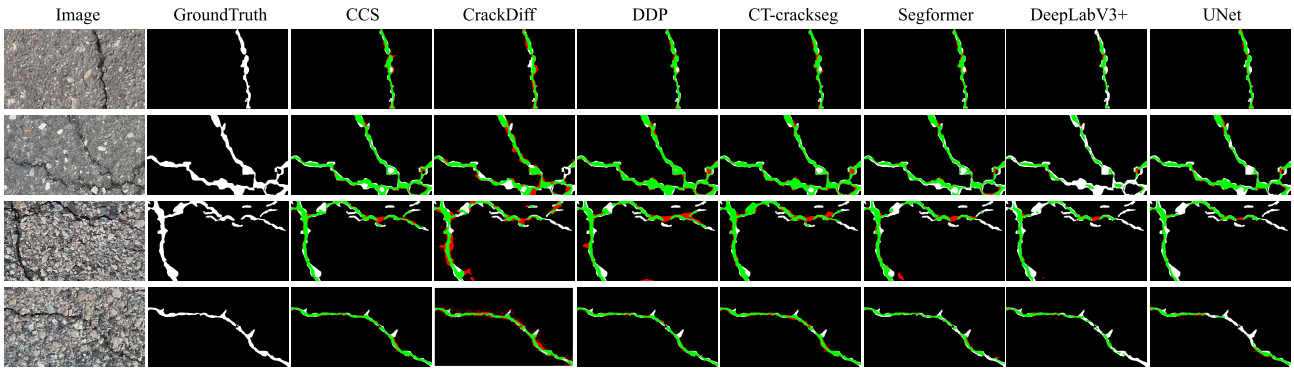

Fig. 7. Visualization results on Crack500. Green pixels represent correctly detected pixels, red indicates false positive pixels, and white represents false negative pixels.

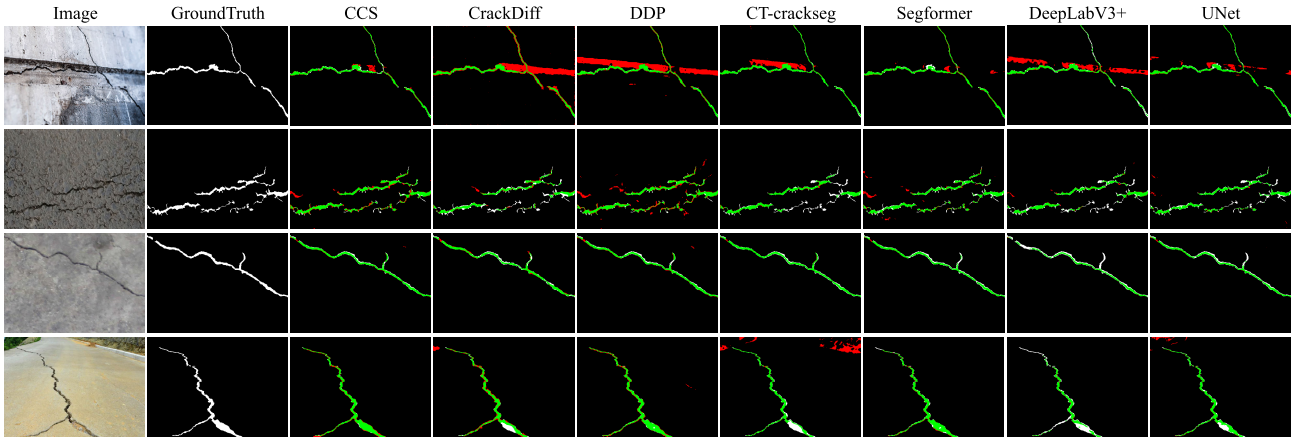

Fig. 8. Visualization results on DeepCrack. Green pixels represent correctly detected pixels, red indicates false positive pixels, and white represents false negative pixels.

of DeepLabV3+ suggest insufficient performance in terms of handling tasks with highly imbalanced data. Despite the presence of clear features, the model struggles with accurately performing classification.

Fig. 8 presents a visualization of predictions derived from all methods. The visualization shows that CCS effectively eliminates most of the interference, producing accurate predictions of continuous and dense cracks. CCS prediction results are comprehensive, showing clear advantages from both fine-grained and global continuity perspectives. Although DDP also performs well, it lacks sufficient interference suppression capabilities, leading to noticeable prediction errors. CrackDiff predictions are competitive but lack a fine-grained learning capacity, reducing the sensitivity to small cracks. CT-crackseg's predictions include more errors, indicating that the model may be overoptimized for cracks, neglecting the influences of other image features.

3) Results on CFD: Table III presents the evaluation results for the CFD dataset. The cracks contained in the CFD dataset are elongated and shallow, with a limited number of images and significant background noise, which creates considerable challenges for the models. CCS achieves the highest mIoU score among the models. DeepLabV3+, U-Net, CrackFormer, CT-crackseg, and CrackDiff all present low Recall values (49.18, 38.61, 58.9, 66.4, and 62.2), indicating a limited ability to detect subtle cracks. Although SegFormer and DDP achieve high Recall values (79.06 and 77.96, respectively), their low

TABLE II

EVALUATION RESULTS OF DIFFERENT MODELS ON DEEPCRACK

Model	Precision	Recall	F1	mIoU
U-Net	94.1	78.86	85.81	87.02
DeepLabV3+	**95.53**	63.2	76.08	79.86
SegFormer(b0)	91.15	81.05	85.8	87.15
CrackFormer	88.32	78.11	82.90	85.46
CT-crackseg	89.76	86.87	88.29	88.39
CrackDiff	91.93	79.45	84.06	86.24
DDP(Swin)	86.35	85.82	86.08	87.2
CCS(ours)	90.51	**86.48**	**88.45**	**89.17**

Precision scores (68.58 and 69.82, respectively) suggest the potential misclassification of indistinct background features as cracks.

Fig. 9 shows a visualization of the predictions derived from all the methods. Using diffusion models, CCS predictions displays the most detailed and uniform cracks in its predictions and even predicts some unlabeled cracks, explaining CCS's lower precision score. CrackDiff are also competitive; however, while the predicted cracks are detailed, some global minor cracks remain omitted. CT-crackseg's boundary masks depend heavily on the quality of the input labels; labeling errors or omissions cause the predictions to focus on labeled cracks, thus neglecting unlabeled minor cracks.

4) Results Analysis and Discussion: Based on the evaluations conducted across three datasets, the CCS model achieves the highest Recall, F1-score, and mIoU scores, but its Precision

Authorized licensed use limited to: Civil Aviation University of China. Downloaded on February 07,2026 at 12:25:49 UTC from IEEE Xplore. Restrictions apply.

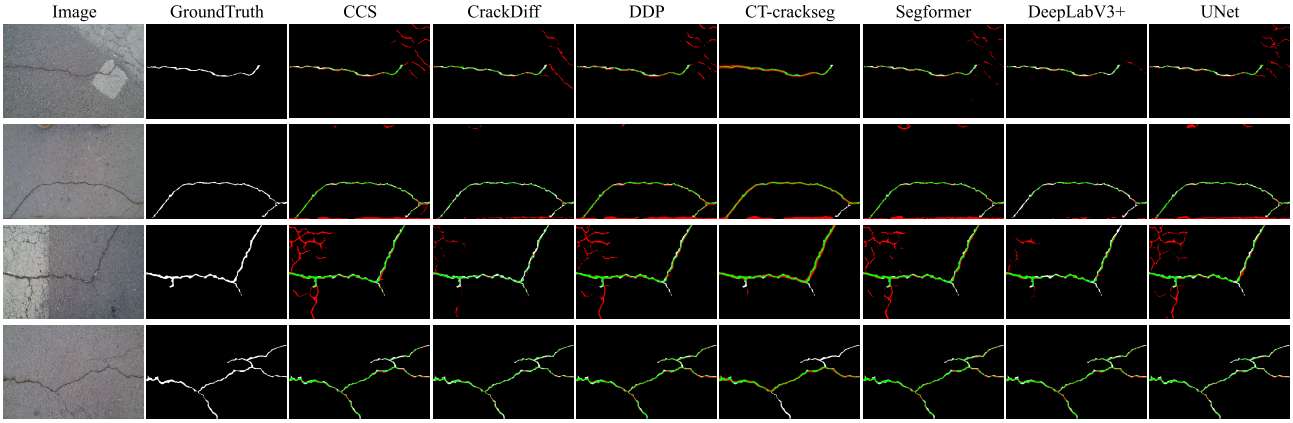

Fig. 9. Visualization results on CFD. Green pixels represent correctly detected pixels, red indicates false positive pixels, and white represents false negative pixels.

TABLE III
EVALUATION RESULTS OF DIFFERENT MODELS ON CFD

Model	Precision	Recall	F1	mIoU
U-Net	76.6	49.18	59.9	70.12
DeepLabV3+	**81.57**	38.61	52.41	66.42
SegFormer(b0)	68.58	79.06	73.45	77.91
CrackFormer	78.9	58.9	67.40	74.20
CT-crackseg	62.8	66.4	64.5	71.7
CrackDiff	78.4	62.2	69.4	75.5
DDP(Swin)	69.82	77.96	73.67	78.08
CCS(ours)	71.7	**80.25**	**75.73**	**79.47**

is relatively low. This result indicates that the CCS model excels at identifying road surface cracks, detecting most actual cracks and handling imbalanced classification tasks. However, the low Precision score of CCS suggests that the model occasionally misclassifies negative samples.

Visual analysis reveals missing details in the dataset labels, leading the model to ignore certain features during training, thereby accelerating its convergence process but causing it to classify these features as negative instances. Utilizing its conditional control mechanism, the CCS model learns shallow, deep, and spatial crack features, effectively enhancing the image details and the resulting segmentation accuracy. This characteristic accounts for the model's lower precision score of the model while also highlighting strengths of CCS in terms of managing complex backgrounds and fine detail.

C. Ablation Study

1) Diffusion Module: We conduct experiments to evaluate the effectiveness of the conditional diffusion models in the crack semantic segmentation task. The backbone of the baseline model uses MiT b0, and its decoding head is the same as that of CCS. Table IV summarizes the evaluation metrics. The CCS model demonstrated a notable segmentation accuracy improvement. Specifically, the CCS model achieves an mIoU of 84.46%, representing a 1.44 percentage point increase over the baseline model. This improvement further confirms the advantages of conditional diffusion models in crack semantic segmentation tasks. The parameter count of the CCS model increases modestly, by 1.86 M, whereas its GFLOPs rise significantly, by 2.86 times. This substantial

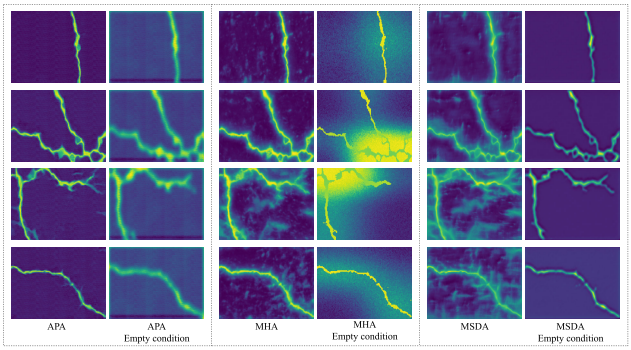

Fig. 10. Visualization results for different attentions.

TABLE IV
EVALUATION RESULTS FOR BASELINE AND CCS

Model	mIoU	GFLOPs	Params
Baseline	83.02	37.97	12.49M
CCS	84.46	108.63	14.35M

TABLE V
EVALUATION RESULTS FOR DIFFERENT ATTENTION MECHANISMS

Attention	mIoU	GFLOPs	Params	FPS
MHA	83.6	206.09	15.76M	0.9
MSDA	83.87	255.64	15.04M	8
APA	84.46	108.63	14.35M	11

increase in computational complexity is due primarily to the CCS model requiring multiple decoder runs during the inference process according to the number of sampling steps.

2) Different Attention Mechanisms: To assess the effectiveness of different attention mechanisms in the map decoder, we conduct comparative experiments involving multi-head attention (MHA), multi-head deformable attention (MSDA), and Aggregated Pixel Attention (APA). We visualize the attention weight heatmaps produced for the final sampling step using Seaborn (Fig. 10), including conditional prediction (left column) and unconditional prediction (right column), providing a more intuitive view of the regions highlighted by each attention mechanism. Table V presents the parameter counts and evaluation metrics of each attention mechanism. The decoder with APA has the lowest parameter count, highest mIoU, and fastest inference speed.

Authorized licensed use limited to: Civil Aviation University of China. Downloaded on February 07,2026 at 12:25:49 UTC from IEEE Xplore. Restrictions apply.

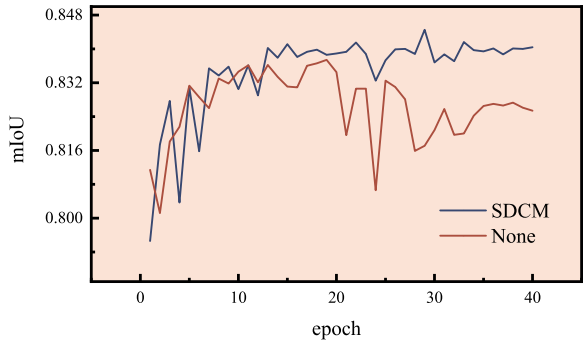

Fig. 11. mIoU of SDCM and None.

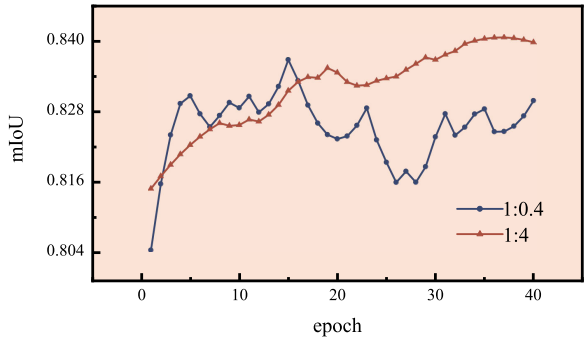

Fig. 12. mIoU of the two proportional loss functions.

Qualitatively, during the inference process, the APA mechanism captures the semantic relationships between local and global crack features, such as boundary details, continuous cracks, and fine crack characteristics. Additionally, APA effectively suppresses noise by focusing its attention weights on crack regions rather than background areas. This focus results in a more accurate and detailed attention heatmap than those generated by other mechanisms. In contrast, MHA and MSDA show substantial deficiencies in their semantic representations, as they produce significant noise. This indicates that the weight allocation of these two attention mechanisms are more sensitive to noise. Increasing the number of sampling steps can mitigate noise to some extent, but it also exponentially increases the computational complexity of the model, reducing its efficiency. For empty condition prediction, the APA mechanism is more inclined to decode intrinsic image features, resulting in a more complete semantic representation in its predictions. In contrast, MHA and MSDA tend to focus on label information, which may lead to the omission of essential semantic details from their predictions.

3) SDCM: This section examines the effect of the SDCM on the performance of the model. The model is evaluated every 2000 iterations. Fig. 11 shows the convergence trends of the model with and without SDCM training. The experimental results indicate that the SDCM effectively assists the model in escaping local optima, leading to stable performance in the later training stages. In contrast, models without the SDCM are prone to sampling drift, tend to become stuck in early local optima, and exhibit performance fluctuations.

The data in Table VI indicate that the SDCM is exclusively used during training and does not affect the parameter count or inference complexity of the model. Consequently, the SDCM mainly impacts the training speed without increasing the storage requirements or inference time of the model. The CCS model incorporating the SDCM achieves an mIoU score of 84.46, indicating a notable improvement over the score of 83.76 obtained without the SDCM. We observe that the CCS model without the SDCM exhibits significant training randomness, although it typically reaches saturation after 30,000 iterations. However, it frequently exhibits performance fluctuations thereafter, affecting the stability of the model.

TABLE VI
EVALUATION RESULTS FOR SDCM

Model	SDCM	mIoU	GFLOPs	Params
CCS		83.76	108.63	14.35M
CCS	✓	84.46	108.63	14.35M

Fig. 13 illustrates the outcomes of the qualitative analysis of the SDCM. In this study, the model weights observed after 2000 training iterations are used to predict the test set, which serves as a reference benchmark. The core mechanism of the diffusion model is to recover predictions from noise, thus inevitably producing prediction noise in the early training stages. This noise can cause the model to converge to a suboptimal solution, thus reducing its prediction accuracy. In this study, labels are depicted as binary images, with their pixel values set to either 0 or 1. The main advantages of the SDCM are its effective ability to suppress the noise generation process and its imposition of constraints on the model predictions. The SDCM enables the rapid adjustment of the map decoder weights by incorporating label information into the prediction process.

4) Sampling Steps: This section examines the effect of the DDIM sampling step size on the resulting crack segmentation performance. Crack segmentation is a binary classification task that distinguishes between cracks and noncracks. Although varying the sampling step size theoretically alters the feature distribution of the input data, the simplicity of this task ensures that the model already effectively distinguishes between the two classes. Thus, adjusting the sampling step size only marginally enhances the representativeness of the data without substantially improving the accuracy of the model. Table VII indicates that increasing the sampling step size from 1 to 2 increases the mIoU from 84.39% to 84.46%, after which the performance plateaus. This suggests that the performance of the model may reach its theoretical upper limit. Notably, GFLOPs nearly linearly increases as the sampling step size increases. This rising computational cost requires careful considerations in practical applications, especially in resource-constrained environments. With a sampling step size of 1, the model faces the challenge of predicting cracks directly from Gaussian noise under empty conditions. In this case, CFG may introduce noise to the prediction results, leading to a decrease in mIoU. Taking these factors into account, a sampling step size of 2 offers an optimal balance between accuracy and computational complexity.

5) Proportions of Two Loss Functions: We investigate the importance of cross-entropy loss and l2 loss that are included in the training process of our model, i.e., the values of λ_1 and λ_2 respectively. Fig. 12 shows the mIoU produced across two

Authorized licensed use limited to: Civil Aviation University of China. Downloaded on February 07,2026 at 12:25:49 UTC from IEEE Xplore. Restrictions apply.

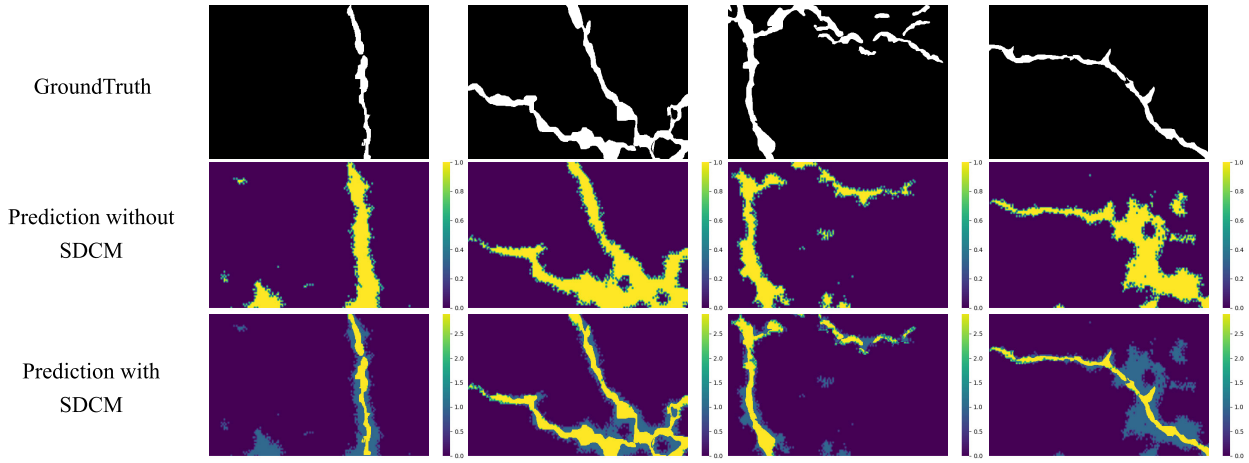

Fig. 13. Visual results of SDCM.

TABLE VII
EVALUATION RESULTS FOR DIFFERENT STEPS

Step	mIoU	GFLOPs	Params	FPS
1	84.39	62.78	14.35M	16.63
2	84.46	108.63	14.35M	9.59
3	84.46	154.47	14.35M	6.69
5	84.46	246.17	14.35M	4.04

settings. Although the primary task is semantic segmentation, the L2 loss of the generative model remains a crucial objective function. Prioritizing cross-entropy loss as the main loss can prevent the model from escaping local optima, limiting its accuracy and causing overfitting. This finding underscores the significance of designing and balancing different loss functions in diffusion-based semantic segmentation tasks. The L2 loss not only helps preserve the generative capacity of the diffusion model but also encourages effective exploration in the feature space, thereby enhancing the overall semantic segmentation performance of the model.

TABLE VIII
MODEL COMPLEXITY AND RUNTIME FRAMES

Model	Img size	Params	GFLOPs	FPS
CCS(b0)	360×640	5.25M	28.82	32
CCS(b3)	360×640	14.35M	108.63	11
UNet	360×640	29.06M	79.12	6
DeepLabV3+	360×640	43.48M	154.88	43
SegFormer	360×640	3.75M	6.17	60
CrackFormer	360×640	7.35M	67.2	15
CT-crackseg	360×640	22.88M	41.62	4
DDP	360×640	39.55M	219.78	13.8

TABLE IX
EVALUATION RESULTS FOR B0 AND B3

Model	CRACK500			DeepCrack			CFD		
	Pr	Re	mIoU	Pr	Re	mIoU	Pr	Re	mIoU
CCS (b0)	81.94	82.84	83.91	87.56	84.09	86.93	70.03	81.28	79.12
CCS (b3)	81.93	84.29	84.46	90.51	86.48	89.17	71.70	80.25	79.47

D. Performance Analysis and Model Complexity

This study investigates the efficacy of diffusion models for completing crack segmentation tasks. We develop a large-parameter model (b3) to comprehensively explore the diffusion parameter space. However, this approach significantly increases the number of learnable parameters and adversely affects the real-time performance of the model. To mitigate these limitations and assess the flexibility of different architectures, we design a lightweight variant (b0). The b0 model reduces the number of required parameters through two primary modifications: (1) decreasing the number of conditional encoder output channels from 256 to 32. (2) reducing the number of decoder layers from 6 to 2. Table VIII presents the parameter counts, GFLOPs, and FPS of all the models. CrackDiff is not discussed in detail here due to its sub-1 FPS performance and multi-step inference requirements. The lightweight strategy reduces the number of model parameters from 14.35M to 5.25M and decreases the computational complexity exhibited at a 640×360 resolution from 108.63 GFLOPs to 28.82 GFLOPs, resulting in an improved inference speed of 32 FPS. While the b0 model exhibits significantly improved computational efficiency and meets the basic requirements of real-time detection, its runtime efficiency remains suboptimal because of the inherent iterative inference mechanisms of diffusion model.

Table IX presents the quantitative evaluation metrics produced for model b0 across three datasets: CRACK500, DeepCrack, and CFD. On the large CRACK500 dataset, model b3 achieves a 1.45% higher Recall than model b0 does, indicating that with sufficient training data, models with more parameters exhibit stronger generative capabilities. For the feature-diverse DeepCrack dataset, model b3 outperforms model b0 across all the evaluation metrics. These results corroborate the superior ability of large-parameter models to process complex feature representations. However, when applied to the CFD dataset, model b3 exhibits overfitting despite the implementation of data augmentation techniques. This performance degradation relative to model b0 can be attributed to the inherently limited feature diversity of this dataset.

Authorized licensed use limited to: Civil Aviation University of China. Downloaded on February 07,2026 at 12:25:49 UTC from IEEE Xplore. Restrictions apply.

V. Conclusion

This research introduces a novel automatic semantic segmentation approach for pavement crack detection, designed to improve the safety of transportation infrastructure and optimize the resource allocation scheme of road maintenance processes. The proposed methodology integrates a redesigned conditional diffusion model with an efficient encoder-decoder architecture. This integration, which is enhanced by CFG, dual optimization training with the SDCM, and a pixel-level attention mechanism, substantially improves both the segmentation accuracy and fine-detail preservation capabilities of the constructed model in crack segmentation tasks. Comprehensive evaluations across three benchmark datasets—CRACK500, DeepCrack, and CFD—validate the exceptional generalizability and advanced spatial feature learning capabilities of the model. The proposed method achieves SOTA performance, with mIoU scores of 84.46%, 89.17%, and 79.47%, respectively, demonstrating robust performance across diverse crack scenarios. Further architectural optimization experiments demonstrate that our lightweight implementation achieves real-time processing capabilities (32 FPS) without significant accuracy degradations. This optimal balance between computational efficiency and segmentation precision establishes a practical framework for deploying diffusion-based models in resource-constrained environments.

In this study, CCS demonstrates strong performance; however, several limitations remain and must be addressed in future work.

1) Significant computational resources are consumed by background pixels, resulting in high computational complexity for the model. Future work will focus on developing efficient foreground-background separation strategies to allocate computational resources more effectively to the foreground pixel calculations.
2) The proposed model employs the most parameter configurations. Although reducing the number of parameters can increase operational efficiency, further research is required to develop more efficient inference methods and mitigate the limitations of the iterative DDIM inference mechanism.
3) Crack annotations are complex and relatively inaccurate. In future work, we aim to enhance the generative capabilities of the proposed diffusion model, develop automated annotation procedures, and employ semisupervised and pseudolabeling methods for model training, thereby improving the efficiency of deployment.
4) Due to the structured design of the model, extending it to additional tasks is both straightforward and feasible. This represents a promising direction for future research.

References

[1] N. Kheradmandi and V. Mehranfar, "A critical review and comparative study on image segmentation-based techniques for pavement crack detection," *Construct. Building Mater.*, vol. 321, Feb. 2022, Art. no. 126162.

[2] Y.-J. Cha, W. Choi, and O. Büyüköztürk, "Deep learning-based crack damage detection using convolutional neural networks," *Comput.-Aided Civil Infrastruct. Eng.*, vol. 32, no. 5, pp. 361–378, May 2017.

[3] Y. Cha, R. Ali, J. S. Lewis, and O. Büyüköztürk, "Deep learning-based structural health monitoring," *Autom. Construct.*, vol. 161, Mar. 2024, Art. no. 105328.

[4] Y.-J. Cha, W. Choi, G. Suh, S. Mahmoudkhani, and O. Büyüköztürk, "Autonomous structural visual inspection using region-based deep learning for detecting multiple damage types," *Comput-Aided Civ. Inf. Eng.*, vol. 33, no. 9, pp. 731–747, 2018.

[5] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assisted Intervent*, Jan. 2015, pp. 234–241.

[6] E. Shelhamer, J. Long, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017.

[7] E. Xie et al., "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 12077–12090.

[8] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 801–818.

[9] W. Choi and Y.-J. Cha, "SDDNet: Real-time crack segmentation," *IEEE Trans. Ind. Electron.*, vol. 67, no. 9, pp. 8016–8025, Sep. 2020.

[10] D. H. Kang and Y.-J. Cha, "Efficient attention-based deep encoder and decoder for automatic crack segmentation," *Struct. Health Monitor.*, vol. 21, no. 5, pp. 2190–2205, Sep. 2022.

[11] F. Liu and L. Wang, "UNet-based model for crack detection integrating visual explanations," *Construct. Building Mater.*, vol. 322, Mar. 2022, Art. no. 126265.

[12] J. Sohl-Dickstein, E. Weiss, N. Maheswaranathan, and S. Ganguli, "Deep unsupervised learning using nonequilibrium thermodynamics," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 2256–2265.

[13] J. Wu, H. Fang, Y. Zhang, Y. Yang, and Y. Xu, "MedSegDiff: Medical image segmentation with diffusion probabilistic model," in *Medical Imaging with Deep Learning*. PMLR, 2024, pp. 1623–1639.

[14] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Proc. NIPS*, vol. 33. Vancouver, BC, Canada: Curran Associates, 2020, pp. 6840–6851.

[15] H. Zhang, N. Chen, M. Li, and S. Mao, "The crack diffusion model: An innovative diffusion-based method for pavement crack detection," *Remote Sens.*, vol. 16, no. 6, p. 986, Mar. 2024.

[16] Z. Lai et al., "Denoising diffusion semantic segmentation with mask prior modeling," 2023, *arXiv:2306.01721*.

[17] F. Liu, W. Ding, Y. Qiao, and L. Wang, "Transfer learning-based encoder–decoder model with visual explanations for infrastructure crack segmentation: New open database and comprehensive evaluation," *Underground Space*, vol. 17, pp. 60–81, Dec. 2023.

[18] J. Hwa, Q. Zhao, A. Lahiri, A. Masood, B. Salimi, and E. Adeli, "Enforcing conditional independence for fair representation learning and causal image generation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2024, pp. 103–112.

[19] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Jun. 2017, pp. 5998–6008.

[20] J. Huyan, W. Li, S. Tighe, Z. Xu, and J. Zhai, "CrackU-net: A novel deep convolutional neural network for pixelwise pavement crack detection," *Struct. Control Health Monitor.*, vol. 27, no. 8, p. 2551, Aug. 2020.

[21] C. Han, T. Ma, J. Huyan, X. Huang, and Y. Zhang, "CrackW-Net: A novel pavement crack image segmentation convolutional neural network," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 11, pp. 22135–22144, Nov. 2022.

[22] J. Chen and Y. He, "A novel U-shaped encoder–decoder network with attention mechanism for detection and evaluation of road cracks at pixel level," *Comput.-Aided Civil Infrastruct. Eng.*, vol. 37, no. 13, pp. 1721–1736, Nov. 2022.

[23] A. Wang, H. Lang, Z. Chen, Y. Peng, S. Ding, and J. J. Lu, "The two-step method of pavement pothole and raveling detection and segmentation based on deep learning," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 6, pp. 5402–5417, Jun. 2024.

[24] Z. Ge, S. Liu, F. Wang, Z. Li, and J. Sun, "YOLOX: Exceeding YOLO series in 2021," 2021, *arXiv:2107.08430*.

[25] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent.*, 2020. Accessed: Mar. 5, 2024.

[26] F. Guo, J. Liu, Q. Xie, and H. Yu, "A two-stage framework for pixel-level pavement surface crack detection," *Eng. Appl. Artif. Intell.*, vol. 133, Jul. 2024, Art. no. 108312.

Authorized licensed use limited to: Civil Aviation University of China. Downloaded on February 07,2026 at 12:25:49 UTC from IEEE Xplore. Restrictions apply.

[27] H. Liu, J. Yang, X. Miao, C. Mertz, and H. Kong, "CrackFormer network for pavement crack segmentation," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 9, pp. 9240–9252, May 2023.

[28] H. Tao, B. Liu, J. Cui, and H. Zhang, "A convolutional-transformer network for crack segmentation with boundary awareness," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2023, pp. 86–90.

[29] Z. Qu, C. Cao, L. Liu, and D.-Y. Zhou, "A deeply supervised convolutional neural network for pavement crack detection with multiscale feature fusion," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 9, pp. 4890–4899, Sep. 2022.

[30] X. Sun, Y. Xie, L. Jiang, Y. Cao, and B. Liu, "DMA-Net: DeepLab with multi-scale attention for pavement crack segmentation," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 10, pp. 18392–18403, Oct. 2022.

[31] D. Baranchuk, I. Rubachev, A. Voynov, V. Khrulkov, and A. Babenko, "Label-efficient semantic segmentation with diffusion models," 2021, *arXiv:2112.03126*.

[32] Y. Ji et al., "DDP: Diffusion model for dense visual prediction," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 21684–21695.

[33] C. Han, H. Yang, T. Ma, S. Wang, C. Zhao, and Y. Yang, "CrackDiffusion: A two-stage semantic segmentation framework for pavement crack combining unsupervised and supervised processes," *Autom. Construct.*, vol. 160, Apr. 2024, Art. no. 105332.

[34] C. Li et al., "CrackCLF: Automatic pavement crack detection based on closed-loop feedback," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 6, pp. 5965–5980, Jun. 2024.

[35] D. Ma, H. Fang, N. Wang, C. Zhang, J. Dong, and H. Hu, "Automatic detection and counting system for pavement cracks based on PCGAN and YOLO-MF," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 11, pp. 22166–22178, Nov. 2022.

[36] R. Ali and Y. Cha, "Attention-based generative adversarial network with internal damage segmentation using thermography," *Autom. Construct.*, vol. 141, Jun. 2022, Art. no. 104412.

[37] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," 2020, *arXiv:2010.02502*.

[38] C. Luo, "Understanding diffusion models: A unified perspective," 2022, *arXiv:2208.11970*.

[39] J. Ho and T. Salimans, "Classifier-free diffusion guidance," 2022, *arXiv:2207.12598*.

[40] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.

[41] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 936–944.

[42] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2999–3007.

[43] T. Amit, T. Shaharbany, E. Nachmani, and L. Wolf, "SegDiff: Image segmentation with diffusion probabilistic models," 2021, *arXiv:2112.00390*.

[44] D. Shi, "TransNeXt: Robust foveal visual perception for vision transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 17773–17783.

[45] M. Dehghani et al., "Scaling vision transformers to 22 billion parameters," in *Proc. Int. Conf. Mach. Learn.*, Jan. 2023, pp. 7480–7512.

[46] J. Su. *Viewing Attention's scale operation through entropy invariance*. Accessed: Dec. 2021. [Online]. Available: https://kexue.fm/archives/8823

[47] Z. Liu et al., "Swin transformer V2: Scaling up capacity and resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 12009–12019.

[48] S. Chen, P. Sun, Y. Song, and P. Luo, "DiffusionDet: Diffusion model for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 19830–19843.

[49] T. Chen, R. Zhang, and G. Hinton, "Analog bits: Generating discrete data using diffusion models with self-conditioning," 2022, *arXiv:2208.04202*.

[50] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," 2019, *arXiv:1711.05101*.

[51] MMSegmentation Contributors. *MMDetection: Open MMLab detection toolbox and benchmark*. Accessed: 2020. [Online]. Available: https://github.com/open-mmlab/mmsegmentation

[52] F. Yang, L. Zhang, S. Yu, D. Prokhorov, X. Mei, and H. Ling, "Feature pyramid and hierarchical boosting network for pavement crack detection," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 4, pp. 1525–1535, Apr. 2020.

[53] Y. Liu, J. Yao, X. Lu, R. Xie, and L. Li, "DeepCrack: A deep hierarchical feature learning architecture for crack segmentation," *Neurocomputing*, vol. 338, pp. 139–153, Apr. 2019.

[54] Y. Shi, L. Cui, Z. Qi, F. Meng, and Z. Chen, "Automatic road crack detection using random structured forests," *IEEE Trans. Intell. Transp. Syst.*, vol. 17, no. 12, pp. 3434–3445, Dec. 2016.

Jihe Xu was born in 2001 in Jiangsu, China. He is currently pursuing the master's degree in transportation from Guangxi University of Science and Technology. His main research interests include deep learning, generative modeling, and multimodal perception.

Chengtao Zhang was born in Jiangsu, China, in 1978. He received the Ph.D. degree in vehicle engineering from China Agricultural University in 2013. He is currently a Graduate Tutor with the School of Mechanical and Automotive Engineering, Guangxi University of Science and Technology, and the Director of the 8th Council of Guangxi Mechanical Engineering Society. His main research interests include vehicle intelligent control technology, in-vehicle embedded systems, ADAS algorithm simulation, and intelligent connected vehicles.

Wei Zeng was born in Guizhou, China, in 1998. He is currently pursuing the master's degree in mechanical engineering with Guangxi University of Science and Technology. His main research interest include deep learning laser radar detection of wheat.

Authorized licensed use limited to: Civil Aviation University of China. Downloaded on February 07,2026 at 12:25:49 UTC from IEEE Xplore. Restrictions apply.